

Task 1: Project Phase-I Summary

This project looks at credit card fraud detection, framed as a binary classification problem: given a set of transaction-level features, predict whether an individual transaction is fraudulent or legitimate. The target variable is Class, where 1 is fraud and 0 is a legitimate transaction.

The dataset used is the Credit Card Fraud Detection dataset, sourced from Kaggle and originally by the Machine Learning Group at Universite Libre de Bruxelles (ULB), covering transactions made by European cardholders over two days in September 2013. The dataset contains 284,807 records across 31 columns: 30 features (Time, V1-V28, and Amount) plus the target Class. V1-V28 are anonymised, PCA-transformed components with no disclosed real-world meaning, while Time and Amount are the only two features in their original form.

Phase 1 saw many key data-quality characteristics. The dataset contained 1,081 duplicate rows, which were removed to prevent data leakage between training and test sets. Amount was severely right-skewed (skewness of 16.979), which was addressed through a log transformation, reducing skewness to 0.161. No missing values, invalid values, or inconsistent entries were found. The most significant characteristic of the dataset is the severe class imbalance. Only 492 of the original 284,807 transactions (0.17%) are fraudulent, a proportion that stayed unchanged after cleaning (473 of 283,726, 0.167%).

This problem is suited to an LCS-based solution for many reasons. LCS systems evolve interpretable, human-readable rule sets rather than opaque decision boundaries, which is needed in a fraud-detection context where decisions may need to be explained to stakeholders such as banks or regulators. In addition, Phase 1's correlation analysis showed that only a few features (V17, V14, V12, V10, V11, V4) had linear signals for predicting fraud, whilst many features contribute little to nothing. That is a pattern LCS's rule discovery mechanism is suited to exploit, since it learns to condition rules on relevant feature ranges rather than needing all 30 features to contribute. The dataset's structure (fully numeric, tabular, binary-labelled) is also compatible with standard LCS implementations such as eLCS.

Task 2: Original LCS System on raw Dataset

Students must run an original LCS system on the raw or original dataset before making improvements. Students are required to:

- Select an appropriate LCS variant, preferably eLCS.
- Use the original LCS code without algorithmic modification.

- Run the original LCS system on the raw or minimally processed dataset.
- Report the selected LCS parameters.
- Record the baseline performance results.
- Explain any minimal processing required to make the dataset compatible with the LCS code.

Task 3: Data Preprocessing and Feature Engineering

Students must improve the quality of the dataset before developing the improved system and make it suitable for working with LCS. Students are required to:

- Address missing values, duplicate records, invalid values, inconsistent categories, and incorrect data types.
- Investigate and handle outliers where appropriate.
- Encode categorical variables and scale, discretise, or transform numerical variables where required.
- Address class imbalance where relevant.
- Apply feature engineering, feature selection, or dimensionality reduction where justified.
- Clearly explain how preprocessing is expected to improve LCS performance.
- Ensure that preprocessing avoids data leakage.

Task 4: Improved LCS-Based System

Students must improve the LCS code, LCS configuration, or machine learning pipeline to develop a novel improved system. Possible improvements include:

- Hyperparameter tuning.
- Improved preprocessing designed for LCS performance.
- Feature selection or dimensionality reduction.
- Rule pruning or rule compaction.
- Modified LCS algorithm.
- Class-imbalance handling.
- Cost-sensitive learning.

Students are required to:

- Clearly identify what was changed compared with the original LCS system.
- Justify why the change may improve prediction accuracy or model quality.
- Provide a brief algorithmic explanation, pseudocode, or flowchart of the improved system.
- Compare the improved system with the original LCS baseline.

Task 5: Experiments

Students must use a suitable and justified validation strategy. Students are required to:

- Use either an *80/20 train-test split* or *k-fold cross-validation*.
- Justify the selected validation approach.

Students must evaluate results using appropriate metrics and statistical tests.

Students are required to:

- Select evaluation metrics suitable for the selected problem.
- For classification, consider accuracy, precision, recall, F1-score, balanced accuracy, ROC-AUC, PR-AUC, and confusion matrix.
- For regression, consider MAE, MSE, RMSE, MAPE, R^2 , and residual analysis.
- Justify the selected metrics.
- Apply at least one suitable statistical test to compare model performance.
- Interpret whether performance differences are statistically significant.

Task 6: Model Comparison

Students must compare the proposed improved LCS-based system with the original LCS and at least three conventional machine learning models. The comparison must include:

- Original LCS on the raw or minimally processed dataset.
- Original LCS on the preprocessed dataset.
- Improved LCS-based system on the preprocessed and/or feature-engineered dataset.
- At least three conventional non-deep-learning machine learning models.

Conventional models may include Random Forest, Support Vector Machine, Decision Tree, Logistic Regression, Naive Bayes, k-Nearest Neighbours, Gaussian Process, or other appropriate non-deep-learning models. Deep-learning models are not permitted for the required model comparison.

Task 7: Interpretation of Results

Students must include a separate section on result interpretation and Explainable AI. Students are required to:

- Interpret the main findings from the model comparison.
- Explain important LCS rules, feature conditions, or classifier patterns.
- Use at least *three sample LCS rules/classifiers* to discuss the decision-making process.
- Discuss how LCS rule-based outputs support interpretability.

- Discuss whether the explanations are trustworthy and practically meaningful.

Task 8: Discussion, Limitations, and Future Work

Students must critically discuss the results and the responsible use of the developed system. Students are required to discuss:

- Reasons for performance improvement or lack of improvement.
- Practical meaning of the results, not only numerical performance.
- Limitations due to data quality, dataset size, class imbalance, noise, missingness, or feature relevance.
- Future work, such as improved LCS variants, better preprocessing, external validation, or domain expert review.

Deliverables

Each group must submit:

- Final report: 5,000--6,500 words, excluding references and appendices.
Source code or Jupyter Notebook.
GitHub repository link and commit-history evidence.
Contribution statement, including a brief individual reflection from each group member.
Cleaned and feature-engineered dataset, or clear dataset reproduction instructions.
Turnitin similarity report with a similarity score not exceeding 20%